AER309 Basic Thermodynamics

Term Test 1 practice A

Chapters 1 to 5 | 80 minutes | 100 marks

Closed book, with a calculator and one letter-sized, one-sided formula sheet. Show all working; no marks are given for an answer alone. Question 5 needs Table A-12 (saturated refrigerant-134a, pressure table).

Constants. Air: $R = 0.287$, $c_p = 1.005$, $c_v = 0.718$ kJ/(kg·K). Liquid water: $c = 4.18$ kJ/(kg·K). Mercury: $\rho = 13\,600$ kg/m^3. $\sigma = 5.67 \times 10^{-8}$ W/(m^2·K^4). $g = 9.81$ m/s^2.

Each question is on its own page with space for your working; the worked solution starts on the next page. Answers are given to four significant figures. Sign convention: $Q > 0$ into the system, $W > 0$ out of it.

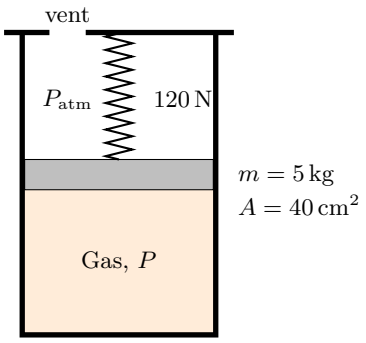
Question 1 [10 marks]

A gas is held in a vertical, frictionless piston-cylinder device. The piston has a mass of 5 kg and a cross-sectional area of 40 cm^2. A compressed spring above the piston pushes down on it with a force of 120 N. The space above the piston, which holds the spring, is vented to the atmosphere. A mercury barometer in the same room reads 740 mm.

(a) Find the atmospheric pressure in kPa.

(b) Find the gage pressure and the absolute pressure of the gas.

(c) A second tank in the room has a gauge that reads 30 kPa of vacuum. Find the absolute pressure in that tank.



Solution to Question 1

Sign convention Upward forces on the piston are positive.

Given $m = 5\,\text{kg}$, $A = 40\,\text{cm}^2 = 40\times10^{-4}\,\text{m}^2$, $F_\text{s} = 120\,\text{N}$ (downward), $h_\text{Hg} = 0.740\,\text{m}$, $P_\text{vac} = 30\,\text{kPa}$.

Find (P_atm, P_gage, P, P_tank)

Assumptions The piston is at rest and frictionless. The space that holds the spring is open to the atmosphere.

Analysis *(a) Barometer.* The mercury column is held up by the atmosphere:

$$P_\text{atm} = \rho_\text{Hg}\, g\, h_\text{Hg} = \left(13\,600\,\text{kg/m}^3\right)\left(9.81\,\text{m/s}^2\right)(0.740\,\text{m}) = 98\,728\,\text{Pa} = 98.728\,\text{kPa}.$$

(b) Vertical forces on the piston. The gas pushes up. The atmosphere, the weight and the spring push down.

$$\sum F_y = 0 = PA - P_\text{atm}A - mg - F_\text{s} \quad\Longrightarrow\quad P = P_\text{atm} + \frac{mg + F_\text{s}}{A}.$$

The second term is the gage pressure, because gage pressure is measured from P_atm:

$$P_\text{gage} = \frac{mg + F_\text{s}}{A} = \frac{(5\,\text{kg})(9.81\,\text{m/s}^2) + 120\,\text{N}}{40\times10^{-4}\,\text{m}^2} = \frac{169.05\,\text{N}}{40\times10^{-4}\,\text{m}^2} = 42\,263\,\text{Pa} = 42.263\,\text{kPa},$$
$$P = P_\text{atm} + P_\text{gage} = 98.728 + 42.263 = 140.99\,\text{kPa}.$$

(c) Vacuum gauge. A vacuum reading is how far the pressure sits below atmospheric:

$$P_\text{tank} = P_\text{atm} - P_\text{vac} = 98.728 - 30 = 68.728\,\text{kPa}.$$

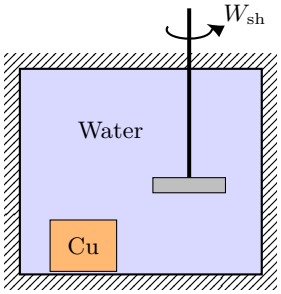
(a) $P_\text{atm} = 98.73\,\text{kPa}$	(b) $P_\text{gage} = 42.26\,\text{kPa}$, $P = 141.0\,\text{kPa}$ absolute	(c) $P_\text{tank} = 68.73\,\text{kPa}$ absolute

Discussion The area has to be in m^2 before dividing: $40\,\text{cm}^2 = 40\times10^{-4}\,\text{m}^2$, not $0.4\,\text{m}^2$. Property tables and the ideal-gas law always take the absolute pressure, never the gage value.

Question 2 [15 marks]

A well-insulated rigid tank holds 10 kg of liquid water at 20 °C. A 3 kg copper block at 200 °C is dropped into the water. At the same moment a paddle wheel in the tank is switched on. Its shaft turns at 240 rpm against a torque of 5 N m for 15 min, and then it stops.
Take $c = 0.386\,\text{kJ/(kg K)}$ for copper.

(a) Find the shaft work done on the contents of the tank.
(b) Find the final equilibrium temperature.



Solution to Question 2

Sign convention $Q > 0$ into the system, $W > 0$ out of the system. The paddle does work on the system, so $W = -W_{\text{sh,in}}$.

Given $m_{\text{w}} = 10\,\text{kg}$, $T_{\text{w}} = 20\,^\circ\text{C}$, $m_{\text{c}} = 3\,\text{kg}$, $T_{\text{c}} = 200\,^\circ\text{C}$, $c_{\text{w}} = 4.18\,\text{kJ/(kg K)}$, $c_{\text{c}} = 0.386\,\text{kJ/(kg K)}$, $\dot{n} = 240\,\text{rpm}$, $\tau = 5\,\text{N m}$, $\Delta t = 15\,\text{min}$.

Find $(W_{\text{sh,in}},\, T_{\text{f}})$

Strategy Take the water and the copper together as one closed system. Then the heat that passes between them is internal and drops out of the balance.

Assumptions The tank is insulated, so $Q = 0$. It is rigid, so there is no boundary work. Water and copper are incompressible with constant specific heats. The water stays liquid. Kinetic and potential energy changes are zero.

Analysis *(a) Shaft work.* The shaft speed in revolutions per second is $\dot{n} = 240/60 = 4/\text{s}$, and $\Delta t = 900\,\text{s}$.

$$W_{\text{sh,in}} = 2\pi\,\dot{n}\,\tau\,\Delta t = 2\pi\,(4/\text{s})\,(5\,\text{N m})\,(900\,\text{s}) = 113\,097\,\text{J} = 113.10\,\text{kJ}.$$

(b) Energy balance on water plus copper.

$$Q - W = \Delta U \quad\Longrightarrow\quad 0 + W_{\text{sh,in}} = m_{\text{w}}c_{\text{w}}\,(T_{\text{f}} - T_{\text{w}}) + m_{\text{c}}c_{\text{c}}\,(T_{\text{f}} - T_{\text{c}})\,.$$

Solving for the final temperature,

$$T_{\text{f}} = \frac{W_{\text{sh,in}} + m_{\text{w}}c_{\text{w}}T_{\text{w}} + m_{\text{c}}c_{\text{c}}T_{\text{c}}}{m_{\text{w}}c_{\text{w}} + m_{\text{c}}c_{\text{c}}}.$$

The two heat capacities are $m_{\text{w}}c_{\text{w}} = (10)(4.18) = 41.8\,\text{kJ/K}$ and $m_{\text{c}}c_{\text{c}} = (3)(0.386) = 1.158\,\text{kJ/K}$. Then

$$T_{\text{f}} = \frac{113.10 + (41.8)(20) + (1.158)(200)}{41.8 + 1.158} = \frac{1\,180.70\,\text{kJ}}{42.958\,\text{kJ/K}} = 27.48\,^\circ\text{C}.$$

Temperatures can stay in °C here because only temperature differences appear in the balance.

(a) $W_{\text{sh,in}} = 113.1\,\text{kJ}$, done on the water (b) $T_{\text{f}} = 27.48\,^\circ\text{C}$

Check The water gains $(41.8)(27.485 - 20) = 312.9\,\text{kJ}$. The copper loses $(1.158)(200 - 27.485) = 199.8\,\text{kJ}$. The difference is $113.1\,\text{kJ}$, which is the shaft work. Without the paddle the final temperature would be only $24.85\,^\circ\text{C}$. The assumption that the water stays liquid holds, since $27.48\,^\circ\text{C}$ is far below boiling.

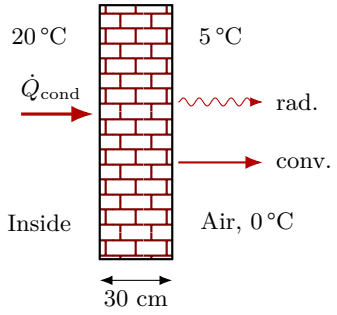
Question 3 [15 marks]

A brick wall is 4 m high, 6 m wide and 30 cm thick, with a thermal conductivity of 0.69 W/(m K). On a winter night its inner surface is at 20 °C and its outer surface is at 5 °C, and conditions are steady. The outer surface has an emissivity of 0.9. The outdoor air and the surrounding surfaces are both at 0 °C.

(a) Find the rate of heat conduction through the wall.

(b) Find the rate at which the outer surface loses heat by radiation.

(c) Find the convection heat transfer coefficient on the outer surface.



Solution to Question 3

Sign convention Heat flow from the inside to the outside is positive.

Given $A = (4)(6) = 24\,\mathrm{m}^2$, $L = 0.30\,\mathrm{m}$, $k = 0.69\,\mathrm{W/(m\,K)}$, $T_1 = 20\,^\circ\mathrm{C}$, $T_\mathrm{s} = 5\,^\circ\mathrm{C}$, $\varepsilon = 0.9$, $T_\infty = T_\mathrm{surr} = 0\,^\circ\mathrm{C}$.

Find $(\dot{Q}_\mathrm{cond},\, \dot{Q}_\mathrm{rad},\, h)$

Strategy At steady state the outer surface stores no energy. Whatever arrives by conduction has to leave by radiation plus convection.

Assumptions Steady state. One-dimensional conduction with constant k. Uniform surface temperatures.

Analysis *(a) Conduction.*

$$\dot{Q}_\mathrm{cond} = kA\,\frac{T_1 - T_\mathrm{s}}{L} = (0.69\,\mathrm{W/(m\,K)})\left(24\,\mathrm{m}^2\right)\frac{(20-5)\,\mathrm{K}}{0.30\,\mathrm{m}} = 828.0\,\mathrm{W}.$$

A difference of $15\,^\circ\mathrm{C}$ is a difference of $15\,\mathrm{K}$, so no conversion is needed here.

(b) Radiation. This one needs absolute temperatures: $T_\mathrm{s} = 278.15\,\mathrm{K}$ and $T_\mathrm{surr} = 273.15\,\mathrm{K}$.

$$\begin{aligned}\dot{Q}_\mathrm{rad} &= \varepsilon\sigma A\left(T_\mathrm{s}^4 - T_\mathrm{surr}^4\right)\\ &= (0.9)\left(5.67\times10^{-8}\,\mathrm{W/(m^2\,K^4)}\right)\left(24\,\mathrm{m}^2\right)\left(278.15^4 - 273.15^4\right)\mathrm{K}^4\\ &= \left(1.2247\times10^{-6}\,\mathrm{W/K^4}\right)\left(5.9857\times10^{9} - 5.5668\times10^{9}\right)\mathrm{K}^4 = 513.07\,\mathrm{W}.\end{aligned}$$

(c) Energy balance on the outer surface.

$$\dot{Q}_\mathrm{cond} = \dot{Q}_\mathrm{rad} + \dot{Q}_\mathrm{conv} \quad\Longrightarrow\quad \dot{Q}_\mathrm{conv} = 828.0 - 513.07 = 314.93\,\mathrm{W}.$$

Then, from $\dot{Q}_\mathrm{conv} = hA\,(T_\mathrm{s} - T_\infty)$,

$$h = \frac{\dot{Q}_\mathrm{conv}}{A\,(T_\mathrm{s} - T_\infty)} = \frac{314.93\,\mathrm{W}}{(24\,\mathrm{m}^2)(5\,\mathrm{K})} = 2.624\,\mathrm{W/(m^2\,K)}.$$

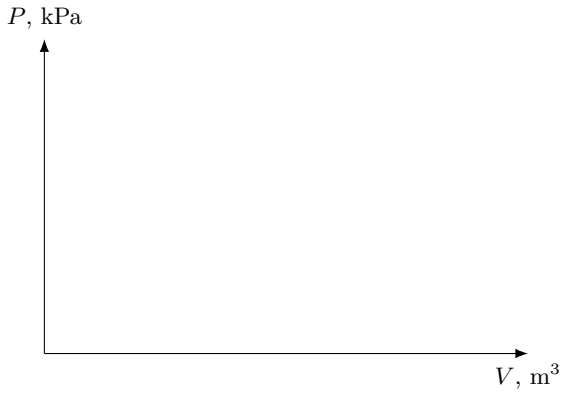
(a) $\dot{Q}_\mathrm{cond} = 828.0\,\mathrm{W}$	(b) $\dot{Q}_\mathrm{rad} = 513.1\,\mathrm{W}$	(c) $h = 2.624\,\mathrm{W/(m^2\,K)}$

Discussion Putting °C into the radiation formula gives $5^4 - 0^4 = 625$ and a heat loss of under a milliwatt, which is the usual way to lose marks here. Radiation carries 62 % of the loss even though the surface is only 5 K warmer than its surroundings. The small h is typical of still air.

Question 4 [20 marks]

A piston-cylinder device contains 0.5 kg of air at 100 kPa and 300 K. The air is compressed slowly to 800 kPa in a polytropic process with $n = 1.3$. Treat air as an ideal gas with constant specific heats.

(a) Find the final temperature and the initial and final volumes.
(b) Find the boundary work, and say whether it is done on or by the air.
(c) Find the heat transfer, and say whether it is into or out of the air.
(d) Sketch the process on a P-V diagram. On the same axes, show where an isothermal compression and an adiabatic ($n = k = 1.4$) compression from the same initial state would end up.



Solution to Question 4

Sign convention $Q > 0$ into the air, $W > 0$ done by the air.

Given $m = 0.5\,\text{kg}$, $P_1 = 100\,\text{kPa}$, $T_1 = 300\,\text{K}$, $P_2 = 800\,\text{kPa}$, $n = 1.3$, $R = 0.287\,\text{kJ/(kg K)}$, $c_v = 0.718\,\text{kJ/(kg K)}$.

Find (T_2, V_1, V_2, W_b, Q)

Assumptions Closed system: the air in the cylinder. Ideal gas with constant specific heats. Quasi-equilibrium process with $PV^n = \text{constant}$. Kinetic and potential energy changes are zero.

Analysis *(a) End states.* For a polytropic process of an ideal gas,

$$T_2 = T_1\left(\frac{P_2}{P_1}\right)^{(n-1)/n} = (300\,\text{K})\left(\frac{800}{100}\right)^{0.3/1.3} = (300\,\text{K})\,(8)^{0.23077} = 484.76\,\text{K}.$$

The volumes follow from $PV = mRT$, using $1\,\text{kJ} = 1\,\text{kPa}\,\text{m}^3$:

$$V_1 = \frac{mRT_1}{P_1} = \frac{(0.5\,\text{kg})(0.287\,\text{kJ/(kg K)})(300\,\text{K})}{100\,\text{kPa}} = 0.4305\,\text{m}^3,$$
$$V_2 = \frac{mRT_2}{P_2} = \frac{(0.5)(0.287)(484.76)}{800}\,\text{m}^3 = 0.08695\,\text{m}^3.$$

(b) Boundary work. With $P = CV^{-n}$,

$$W_\text{b} = \int_1^2 P\,\text{d}V = \frac{P_2V_2 - P_1V_1}{1-n} = \frac{mR\,(T_2 - T_1)}{1-n}.$$

$$W_\text{b} = \frac{(0.5\,\text{kg})(0.287\,\text{kJ/(kg K)})(484.76 - 300)\,\text{K}}{1 - 1.3} = \frac{26.513\,\text{kJ}}{-0.3} = -88.38\,\text{kJ}.$$

The sign is negative, so the work is done on the air.

(c) Energy balance.

$$Q - W_\text{b} = \Delta U \quad\Longrightarrow\quad Q = mc_v\,(T_2 - T_1) + W_\text{b}.$$

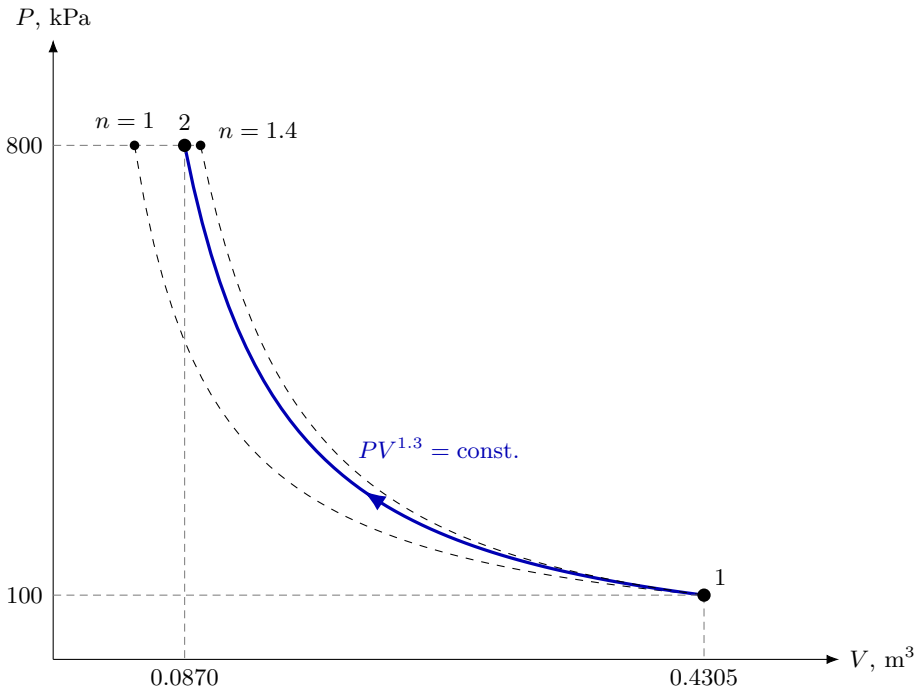
$$\Delta U = (0.5\,\text{kg})(0.718\,\text{kJ/(kg K)})(184.76\,\text{K}) = 66.33\,\text{kJ},$$
$$Q = 66.33 + (-88.38) = -22.05\,\text{kJ}.$$

The sign is negative, so the heat leaves the air.

(a) $T_2 = 484.8\,\text{K}$, $V_1 = 0.4305\,\text{m}^3$, $V_2 = 0.08695\,\text{m}^3$ (b) 88.38 kJ of work done on the air (c) 22.05 kJ of heat out of the air

Discussion Of the 88.38 kJ of work put in, 66.33 kJ stays in the air as internal energy and 22.05 kJ leaves as heat. That fits $n = 1.3$, which lies between isothermal ($n = 1$, all the work leaves as heat) and adiabatic ($n = 1.4$, none of it does). The diagram is on the next page.

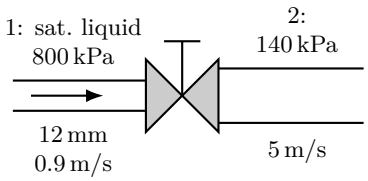
P-V diagram for Question 4 (drawn to scale)



The process runs from state 1 up and to the left. All three curves start at state 1 and stop at 800 kPa. The larger the exponent, the steeper the curve, so the adiabatic path reaches 800 kPa at the largest volume (0.0975 m^3) and the isothermal path at the smallest (0.0538 m^3). The area under the $n = 1.3$ curve, down to the V axis, is the 88.38 kJ of boundary work.

Question 5 [20 marks]

Refrigerant-134a enters the throttling valve of a refrigerator as saturated liquid at 800 kPa and leaves at 140 kPa. The inlet tube has an inside diameter of 12 mm and the liquid moves through it at 0.9 m/s. The valve is well insulated.



(a) Find the mass flow rate.
(b) Find the quality of the refrigerant at the exit.
(c) Find the temperature change across the valve.
(d) Find the exit tube diameter that keeps the exit velocity at 5 m/s.

Solution to Question 5

Sign convention $Q > 0$ into the control volume, $W > 0$ out of it. In this question and the next, V is velocity and $\dot{V}$ is volume flow rate.

Given State 1: saturated liquid, $P_1 = 800\,\text{kPa}$, $D_1 = 0.012\,\text{m}$, $V_1 = 0.9\,\text{m/s}$. State 2: $P_2 = 140\,\text{kPa}$, $V_2 = 5\,\text{m/s}$.

Find $(\dot{m},\, x_2,\, T_2 - T_1,\, D_2)$

Assumptions Control volume: the valve, one inlet and one exit. Steady flow. Adiabatic, and a valve has no shaft, so $\dot{Q} = \dot{W} = 0$. Kinetic and potential energy changes are negligible (checked at the end).

Properties From Table A-12:

P (kPa)	T_{sat} (°C)	v_{f} (m^3/kg)	v_{g} (m^3/kg)	h_{f} (kJ/kg)	h_{fg} (kJ/kg)	h_{g} (kJ/kg)
800	31.31	0.0008457	0.025645	95.48	171.86	267.34
140	−18.77	0.0007381	0.14020	27.06	212.13	239.19

State 1 is saturated liquid, so $v_1 = v_{\text{f}} = 0.0008457\,\text{m}^3/\text{kg}$, $h_1 = h_{\text{f}} = 95.48\,\text{kJ/kg}$ and $T_1 = 31.31\,^\circ\text{C}$.

Analysis *(a) Mass balance.*

$$A_1 = \frac{\pi D_1^2}{4} = \frac{\pi(0.012\,\text{m})^2}{4} = 1.1310 \times 10^{-4}\,\text{m}^2,$$

$$\dot{m} = \frac{V_1 A_1}{v_1} = \frac{(0.9\,\text{m/s})(1.1310 \times 10^{-4}\,\text{m}^2)}{0.0008457\,\text{m}^3/\text{kg}} = 0.12036\,\text{kg/s}.$$

(b) Energy balance. With no heat, no work and negligible kinetic and potential energy, the steady-flow balance reduces to

$$h_2 = h_1 = 95.48\,\text{kJ/kg}.$$

At 140 kPa, $h_{\text{f}} < h_2 < h_{\text{g}}$ $(27.06 < 95.48 < 239.19)$, so the exit is a saturated mixture with

$$x_2 = \frac{h_2 - h_{\text{f}}}{h_{\text{fg}}} = \frac{95.48 - 27.06}{212.13} = 0.32254.$$

(c) Temperature change. A saturated mixture sits at the saturation temperature for its pressure, so $T_2 = -18.77\,^\circ\text{C}$ and

$$T_2 - T_1 = -18.77 - 31.31 = -50.08\,^\circ\text{C}.$$

(d) Exit tube. The exit specific volume is

$$v_2 = v_{\text{f}} + x_2\,(v_{\text{g}} - v_{\text{f}}) = 0.0007381 + (0.32254)(0.14020 - 0.0007381) = 0.045720\,\text{m}^3/\text{kg}.$$

The mass flow rate is the same at the exit, so $\dot{m} = V_2 A_2 / v_2$ and

$$A_2 = \frac{\dot{m} v_2}{V_2} = \frac{(0.12036\,\text{kg/s})(0.045720\,\text{m}^3/\text{kg})}{5\,\text{m/s}} = 1.1006 \times 10^{-3}\,\text{m}^2,$$

$$D_2 = \sqrt{\frac{4A_2}{\pi}} = 0.03743\,\text{m}.$$

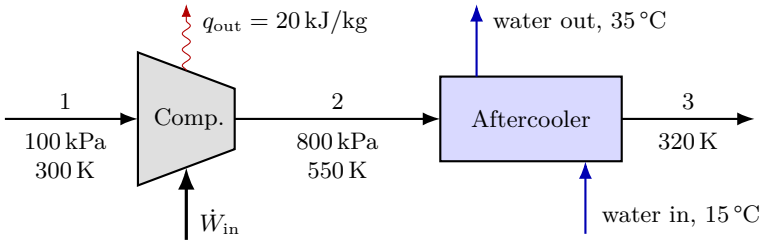
(a) $\dot{m} = 0.1204\,\text{kg/s}$ (b) $x_2 = 0.3225$ (c) the temperature drops by $50.08\,^\circ\text{C}$, from $31.31\,^\circ\text{C}$ to $-18.77\,^\circ\text{C}$ (d) $D_2 = 37.43\,\text{mm}$

Check The exit kinetic energy is $V_2^2/2 = 25/2 = 12.5\,\text{J/kg} = 0.0125\,\text{kJ/kg}$, which is nothing next to $h = 95.48\,\text{kJ/kg}$, so leaving it out was fair. The specific volume grows by a factor of 54 across the valve, which is why the exit tube has to be about three times wider. In the same 12 mm tube the exit velocity would be 48.65 m/s.

Question 6 [20 marks]

Air enters a compressor at 100 kPa and 300 K with a volume flow rate of 1.2 m^3/s. It leaves at 800 kPa and 550 K. The compressor casing loses 20 kJ of heat to the room for every kilogram of air that passes through. The hot air then goes through an aftercooler, a heat exchanger in which cooling water brings the air down to 320 K at constant pressure. The water enters at 15 °C and leaves at 35 °C. The aftercooler is insulated on the outside. Use constant specific heats and neglect kinetic and potential energy changes.

(a) Find the mass flow rate of air.
(b) Find the power input to the compressor.
(c) Find the mass flow rate of cooling water.



Solution to Question 6

Sign convention Each term is written as a positive quantity with its direction in the subscript ($\dot{W}_{\text{in}}$, $\dot{Q}_{\text{out}}$), and the balance is written as energy in equals energy out.

Given $P_1 = 100\,\text{kPa}$, $T_1 = 300\,\text{K}$, $\dot{V}_1 = 1.2\,\text{m}^3/\text{s}$, $P_2 = 800\,\text{kPa}$, $T_2 = 550\,\text{K}$, $q_{\text{out}} = 20\,\text{kJ/kg}$, $T_3 = 320\,\text{K}$, $T_{\text{w,in}} = 15\,^\circ\text{C}$, $T_{\text{w,out}} = 35\,^\circ\text{C}$.

Find ($\dot{m}_{\text{a}}$, $\dot{W}_{\text{in}}$, $\dot{m}_{\text{w}}$)

Assumptions Steady flow in both devices. Air is an ideal gas with constant c_p, so $\Delta h = c_p\,\Delta T$. Water is incompressible and its pressure drop is negligible, so $\Delta h = c\,\Delta T$. Kinetic and potential energy changes are negligible. The aftercooler exchanges no heat with the room and has no work.

Analysis *(a) Mass flow rate of air.* The ideal-gas law gives the inlet specific volume:

$$v_1 = \frac{RT_1}{P_1} = \frac{(0.287\,\text{kJ/(kg K)})(300\,\text{K})}{100\,\text{kPa}} = 0.8610\,\text{m}^3/\text{kg},$$

$$\dot{m}_{\text{a}} = \frac{\dot{V}_1}{v_1} = \frac{1.2\,\text{m}^3/\text{s}}{0.8610\,\text{m}^3/\text{kg}} = 1.3937\,\text{kg/s}.$$

(b) Compressor. Control volume: the compressor. Energy in equals energy out:

$$\dot{W}_{\text{in}} + \dot{m}_{\text{a}} h_1 = \dot{Q}_{\text{out}} + \dot{m}_{\text{a}} h_2 \quad\Longrightarrow\quad \dot{W}_{\text{in}} = \dot{m}_{\text{a}}\left[c_p\,(T_2 - T_1) + q_{\text{out}}\right].$$

The work per unit mass is

$$w_{\text{in}} = (1.005\,\text{kJ/(kg K)})(550 - 300)\,\text{K} + 20\,\text{kJ/kg} = 251.25 + 20 = 271.25\,\text{kJ/kg},$$

so

$$\dot{W}_{\text{in}} = (1.3937\,\text{kg/s})(271.25\,\text{kJ/kg}) = 378.0\,\text{kW}.$$

(c) Aftercooler. Control volume: the whole heat exchanger, with both streams inside it. Nothing crosses that boundary as heat or work, so

$$\dot{m}_{\text{a}} h_2 + \dot{m}_{\text{w}} h_{\text{w,in}} = \dot{m}_{\text{a}} h_3 + \dot{m}_{\text{w}} h_{\text{w,out}} \quad\Longrightarrow\quad \dot{m}_{\text{w}} = \frac{\dot{m}_{\text{a}} c_p\,(T_2 - T_3)}{c_{\text{w}}\,(T_{\text{w,out}} - T_{\text{w,in}})}.$$

The heat given up by the air is

$$\dot{m}_{\text{a}} c_p\,(T_2 - T_3) = (1.3937)(1.005)(550 - 320) = 322.16\,\text{kW},$$

and then

$$\dot{m}_{\text{w}} = \frac{322.16\,\text{kW}}{(4.18\,\text{kJ/(kg K)})(35 - 15)\,\text{K}} = 3.854\,\text{kg/s}.$$

(a) $\dot{m}_{\text{a}} = 1.394\,\text{kg/s}$	(b) $\dot{W}_{\text{in}} = 378.0\,\text{kW}$ into the compressor	(c) $\dot{m}_{\text{w}} = 3.854\,\text{kg/s}$

Discussion The heat loss raises the power the compressor needs. The shaft has to supply the enthalpy rise of the air and also whatever leaks out through the casing, so q_{out} is added, not subtracted. For a turbine it goes the other way. The mass flow rate must come from the inlet state, because the $1.2\,\text{m}^3/\text{s}$ was measured there; the volume flow rate at the exit is much smaller even though $\dot{m}_{\text{a}}$ is the same. In part (c) the choice of boundary matters: with both streams inside the control volume the heat passed from air to water never crosses it, and no $\dot{Q}$ appears.